

LECTURE 24 (DECEMBER 3RD)

Corollary. We would like to observe the different channels in Bhabha scattering in the massless limit. Recall that $\bar{\psi}_{L,R} = \bar{\psi}P_{R,L}$. The decomposition of the QED interaction vertex is:

$$\begin{aligned} eA_\mu \bar{\psi} \gamma^\mu \psi &= eA_\mu \bar{\psi} \gamma^\mu (P_L + P_R) \psi = eA_\mu \bar{\psi} \gamma^\mu (P_L P_L \psi + P_R P_R \psi) \\ &= eA_\mu \bar{\psi}_L \gamma^\mu \psi_L + eA_\mu \bar{\psi}_R \gamma^\mu \psi_R \end{aligned}$$

The decomposition of $\mathcal{M}_{e^-e^+ \rightarrow e^-e^+}$ is

$$M_{e^-e^+ \rightarrow e^-e^+} = \frac{e^2(\bar{U}_- \gamma^\mu U_+)(\bar{U}_+ \gamma_\mu U_-)}{s^2} - \frac{e^2(\bar{U}_+ \gamma^\mu U_+)(\bar{U}_- \gamma_\mu U_-)}{t^2}$$

Let's make some comments:

- (i) If you sum all s -channels, we yield

$$2e^4 \frac{t^2 + u^2}{2}$$

as expected.

- (ii) No cross terms between different helicity structures. This can be easily checked through chirality decomposed spin-state sum.
- (iii) Parity invariance of massless QED, and $|\mathcal{M}|^2$ is invariant under $L \leftrightarrow R$.
- (iv) In the CM frame,

$$\begin{cases} e_L^- e_R^+ \rightarrow e_L^- e_R^+ |_s \sim \frac{u^2}{s^2} \sim \cos^4(\theta/2) \\ e_L^- e_R^+ \rightarrow e_R^- e_L^- |_s \sim \frac{t^2}{s^2} \sim \sin^4(\theta/2) \end{cases}$$

which goes to zero as $\theta \rightarrow \pi, 0$ respectively.

LECTURE 25 (DECEMBER 8TH)

Example. Let's finish off Compton scattering!

$$i\mathcal{M}_{e^- \gamma \rightarrow e^- \gamma} = \left(\bar{u}_+^j(q) i e \gamma^\nu \frac{-i}{\cancel{p} + \cancel{K} + m - i\epsilon} i e \gamma^\mu u_+^i(p) + ((K, \mu) \leftrightarrow -K', \nu) \right) \epsilon_\mu^r(K) \epsilon_\nu^s(K')^*$$

in a expression like the first term, we'd like the operator to be in the numerator, and

$$i\mathcal{M} = i e^2 \left(\frac{\bar{u}_+^j(q) \gamma^\nu (\cancel{p} + \cancel{K} - m) \gamma^\mu u_+^i(p)}{-2p \cdot K} + ((K, \mu) \leftrightarrow (-K', \nu)) \right) \epsilon_\mu^r(K) \epsilon_\nu^s(K')^*$$

multiplying both the numerator and denominator by:

$$\frac{(\cancel{p+K}) - m}{(\cancel{p+K}) - m}$$

Using the two identities

$$(\not{p} \pm m)u_{\pm}(p) = 0 \quad \& \quad p_{\nu}(-\gamma^{\mu}\gamma^{\nu} - 2\eta^{\mu\nu}) = -\gamma^{\mu}\not{p} - 2p^{\mu}$$

we have

$$ie^2 \left(\frac{\bar{u}_+^j(q)(\gamma^{\nu}K^{\mu} - 2\gamma^{\nu}p^{\mu})u_+^j(p)}{-2p \cdot K} + \dots \right) \epsilon_{\mu}^r(K) \epsilon_{\nu}^s(K')^*$$

The scattering amplitude square correspondingly becomes:

$$|\mathcal{M}|^2 = e^4 \left(\frac{\bar{u}_+^i(q)(\gamma^{\nu}K^{\mu} - 2\gamma^{\nu}p^{\mu})u_+^j(p)}{2p \cdot K} + \dots \right) \left(\frac{\bar{u}_+^i(p)(\gamma^{\rho}K^{\sigma} - 2\gamma^{\sigma}p^{\rho})u_+^j(q)}{2p \cdot K} + \dots \right) \\ \times \epsilon_{\mu}^r(K) \epsilon_{\nu}^s(K')^* \epsilon_{\rho}^r(K)^* \epsilon_{\sigma}^s(K')$$

When we want to do the spin polarisation state sum, we use the identities:

$$\sum_{i=1}^2 u_+^i(p) \bar{u}_+^i(p) = -(\not{p} - m)$$

and

$$\sum_{r=1}^2 \epsilon_{\mu}^r(p)^* \epsilon_{\nu}^r(p) = \eta_{\mu\nu} - \frac{p_{\mu}p_{\nu} + (p_{\mu}n_{\nu} + p_{\nu}n_{\mu})(p \cdot n)}{(p \cdot n)^2}$$

This would become a disaster. Here, the saviour of QED is the Ward identity:

$$\mathcal{M} = \mathcal{M}^{\mu} \epsilon_{\mu}^r(p) \implies \mathcal{M}^{\mu} p_{\mu} = 0$$

and this begs us to express all ϵ 's as η 's.

Corollary. (Initial-averaged and final-summed $|\mathcal{M}|^2$)

$$\begin{aligned}
\sum |\mathcal{M}|^2 &= \frac{1}{2} \sum_i \frac{1}{2} \sum_r \sum_j \sum_s |\mathcal{M}|^2 \\
&= \frac{e^4}{4} \text{Tr} \left((\not{q} - m) \frac{\gamma^\nu \not{K} \gamma^\mu - 2\gamma^\nu p^\mu}{2p \cdot K} (\not{p} - m) \frac{\gamma_\mu \not{K} \gamma_\nu - 2\gamma_\nu p_\mu}{2p \cdot K} + \dots \right) \\
&= \frac{e^4}{16} \text{Tr} \left(\frac{\gamma_\nu (\not{q} - m) (\gamma^\mu \not{K} \gamma^\mu - 2\gamma^\nu p^\mu) (\not{p} - m) (\gamma_\mu \not{K} - 2p_\mu)}{(p \cdot K)^2} \right) \\
&= \frac{e^4}{8} \text{Tr} \left(\frac{(\not{q} + 2m)(\not{K} \gamma^\mu - 2p^\mu) (\not{p} - m) (\gamma_\mu \not{K} - 2p_\mu)}{(p \cdot K)^2} \right) \\
&= \frac{e^4}{8} \frac{\text{Tr}[(\not{q} + 2m)(\not{K}(2\not{p} + 4m)\not{K} - 2\not{K}\not{p}(\not{p} - m) - 2(\not{p} - m)\not{p}\not{K} - 4m^2(\not{p} - m))]}{(p \cdot K^2)} \\
&= e^4 \left(\frac{2(q \cdot K)(p \cdot K) + 2m^2 q \cdot (p + K) + 4m^2(m^2 - p \cdot K)}{(p \cdot K)^2} \right) + \dots \\
&= 2e^4 \left(\frac{p \cdot K'}{p \cdot K} + \frac{p \cdot K}{p \cdot K'} - 2m^2 \left(\frac{1}{p \cdot K} - \frac{1}{p \cdot K'} \right) + m^4 \left(\frac{1}{p \cdot K} - \frac{1}{p \cdot K'} \right)^2 \right)
\end{aligned}$$

Corollary. What are the physical observables? We need to partial integrate $d\sigma$ in the LAB frame,

$$\begin{aligned}
\sigma &= \int \frac{d^3 \vec{\omega}'}{(2\pi)^3 \cdot 2\omega'} \frac{d^3 \vec{q}'}{(2\pi)^3 \cdot 2E_q} \frac{1}{2m} \frac{1}{2\omega} (2\pi)^4 \delta^{(4)}(p + K - q - K') \cdot |\mathcal{M}|^2 \\
&= \int \frac{\omega'^2 d\omega' d\Omega}{(2\pi)^3 \cdot 2\omega'} \frac{1}{2E_q} \frac{1}{2m} \frac{1}{2\omega} (2\pi) \delta^{(4)}(m + \omega - E_q - \omega') \cdot |\mathcal{M}|^2
\end{aligned}$$

we now want to substitute

$$E_q = \sqrt{\vec{q}^2 + m^2} = \sqrt{\omega^2 + \omega'^2 - 2\omega\omega' \cos \theta + m^2}$$

and

$$\begin{aligned}
&= \int d\Omega \frac{1}{(2\pi)^2} \frac{\omega'}{16E_q m \omega} \frac{|\mathcal{M}|^2}{\left| 1 + \frac{\omega' - \omega \cos \theta}{E_q} \right|} \Big|_{m+\omega=E_q+\omega'} \\
&= \int d\Omega \frac{1}{(2\pi)^2} \frac{\omega'}{16m\omega} \frac{1}{m + \omega(1 - \cos \theta)} |\mathcal{M}|^2 \\
&= \int d\Omega \left(\frac{\omega'}{8\pi\omega} \right)^2 \cdot |\mathcal{M}|^2
\end{aligned}$$

Now the differential cross section is finally:

$$\begin{aligned}
\left(\frac{d\sigma}{d\Omega} \right)_{\text{LAB}} &= \left(\frac{\omega'}{8\pi m \omega} \right)^2 \times 2e^4 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - 2m \left(\frac{1}{\omega'} - \frac{1}{\omega} \right) + m^2 \left(\frac{1}{\omega'} - \frac{1}{\omega} \right)^2 \right) \\
&= \left(\frac{\omega'}{8\pi m \omega} \right)^2 \times 2e^4 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right)
\end{aligned}$$

Some comments:

- (i) This is called the (spin-averaged) Klein-Nishina formula.
- (ii) Reproduces the classical Thompson scattering cross section in the low energy limit ($\omega \ll m$)
- (iii)